

EE 467 Project Proposal

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Cybersecurity Application Area

For my final project, I've decided to focus on the area of network intrusion detection. I will try to identify malicious network traces that are nested within large amounts of benign data. This is especially important because some of these network intrusions are considered "zero day" attacks, meaning that the intrusion is deployed so early that the victim company or organization has had no time to react. Thus, tracking down and identifying these intrusions is very valuable.

Dataset Selection

For this application, I've chosen the [CIC-IDS2017](#) dataset from the list of provided datasets. I have chosen this dataset because it contains an extensive amount of network connection data, both malicious and benign. There are over 80 features related to each network connection as well as labels describing the specific malicious connection type. Because this is a dataset that contains so many various network connections collected over a period of time, one limitation is that there is much more benign data than malicious data. Like in the labs, I will need to address the limitations of data imbalance as well as duplicate values through rebalancing and cleaning.

Data Type Characterization

The chosen dataset is structured, meaning it has defined features provided. It is also already labeled with whether each network connection is benign or malicious, as well as what type of malicious connection it is. The dataset additionally contains a mixed variety of features. Most of the features are quantitative, such as packet lengths and times, but some are qualitative, such as IP and port information, as well as the malicious/benign type labels themselves. The dataset is time-dependent because it was collected over the span of five days.

Data Science / Machine Learning Problem Definition

The task type for this project is classification; either binary classification between benign and malicious network connections, or multiclass classification between benign and the various malicious attack types. The inputs will be a subset of the 80+ network metrics provided in the dataset, and the output will be the predicted class as described previously. A successful outcome will include a high detection rate with minimal false positives and false negatives. From what we have learned in class, F-1 score seems to be the best metric for measuring success considering the imbalance state of the dataset, however this choice may change as I progress.

Why Data Science / Machine Learning?

An ML-based approach is appropriate for this kind of cybersecurity problem because traditional network firewall solutions are not always able to detect all kinds of malicious network traffic or intrusions. Especially with the rapidly evolving state of bad actors, employing more robust detection solutions such as those involving machine learning may be necessary. ML is also well suited for this problem because it can learn and adapt to patterns in the presented data, even when it is increasingly large, allowing it to spot malicious attacks that may easily blend in with benign data.

Planned Approaches

Given the limitations discussed above, some of the pre-processing steps I will need to take include removing duplicate values and handling and missing values, normalizing value ranges as needed, handling data imbalances potentially using SMOTE, dropping unnecessary qualitative

features, as well as dimension reduction to narrow down the most important or impactful features.

For supervised learning, I plan to try using either the Random Forest model or the XGBoost model, which construct a multitude of decision trees. For unsupervised learning, I plan to try the Isolation Forest model, which from my understanding, seems to focus on isolating anomalies in data instead of separating normal points into classes. I am unsure about using any hybrid options but I may look into them if time permits.

Evaluation Plan

As previously mentioned, I will use F-1 score to report success rates due to the heavy imbalance of data in the network connection dataset. I will also examine the results of precision and recall scores for each predicted class. I will use a time-based train/test split, potentially training on the earlier data recorded during the first few days, and testing on the later data recorded during the last few days of the recording time period. This is as opposed to randomly sampling a 80/20 split from the data, which may disregard time as a factor.

I believe that F-1 score and potentially precision will be the primary metric in this project. Since false positives can be so detrimental, it is important to weigh them significantly, which I think these metrics do a better job of. Recall and accuracy will be the supporting metrics, as they don't do nearly as good of a job. The combination of all these metrics should hopefully give a good idea of how successfully the chosen model(s) are able to detect malicious network attacks.

Team Plan

As I am working on my own, I will be responsible for all relevant tasks, including but not limited to cleaning, feature engineering, model curation and tuning, evaluation, and reporting.